MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
B.Tech. III Semester End Examinations
Model Question Paper(SET-1)

MR-25

Course Title: OOP Through Java

Time: 3 hours Max. Marks : 60

Note: Answer ALL Questions

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M
1. a)	What is abstraction? How does it help in coping with complexity?	1
1. b)	What is the use of the this keyword?	1
1. c)	List any two benefits and two costs of inheritance.	1
1. d)	What is the use of the final keyword with a method and a class?	1
1. e)	Differentiate between throw and throws.	1
1. f)	What is thread synchronization?	1
1. g)	List any four event classes in Java.	1
1. h)	Write the purpose of the CardLayout manager?	1
1. i)	What is an ImageIcon?	1
1. j)	Differentiate between a component and a container in Swing.	1

Part-B (5 x 10=50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
	Unit-I				
2. a)	Demonstrate the Java buzzwords.	5	1	1	1,2
2. b)	Write a program to demonstrate constructor overloading and method overloading.	5	2	1	1,2
	OR				
2. c)	Explain parameter passing in Java with an example. Is Java call by value or call by reference?	5	2	1	1,2
2. d)	Write about nested and inner classes with an example.	5	1	1	1,2
	Unit-II				
3. a)	Explain method overriding with a program. Describe the uses of the super keyword.	5	2	2	1,2
3. b)	Explain how to create and access a package. What is CLASSPATH?	5	2	2	1,2
	OR				
3. c)	Define an abstract class. Write a program with an abstract class Shape and subclasses Circle and Rectangle that compute area.	5	1	2	1,2
3. d)	Does Java Support Multiple Inheritance.Justify	5	4	2	1,2
	Unit-III				
4. a)	Discuss try, catch, throw, throws and finally with a program.	5	3	3	1,2
4. b)	Write a program to create a user-defined exception InvalidAgeException and use it.	5	4	3	1,2
	OR				
4. c)	Illustrate the life cycle of a thread with a neat diagram.	5	2	3	1,2
4. d)	Write a program to create two threads by extending the Thread class and set different priorities.	5	4	3	1,2
	Unit-IV				
5. a)	Explain the ArrayList, HashMap and StringTokenizer classes of java.util with an example.	5	4	4	1,2
5. b)	How mouse events are handled in Java.	5	1	4	1,2
	OR				
5. c)	Summarize the Border, Grid and Flow layout managers with examples.	5	2	4	1,2
5. d)	Explain byte streams and character streams. Write a program to copy one file to another.	5	2	4	1,2
	Unit-V				
6. a)	Write a Swing program that creates a JFrame with a JLabel, a JTextField and a JButton; on clicking the button, display a greeting with the entered name.	5	4	5	1,2
6. b)	Demonstrate MVC architecture in Swing.	5	2	5	1,2
	OR				
6. c)	Write a Swing program using JRadioButton and JCheckBox to select gender and hobbies and display the selection.	5	4	5	1,2
6. d)	Explain JTabbedPane and JScrollPane with an example.	5	2	5	1,2